

E-NOSE & GRAPH NEURAL NETWORKS FOR MOLECULAR PROPERTY PREDICTION

INTERNSHIP PRESENTATION - FIRST WEEK

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1 Problem Statement

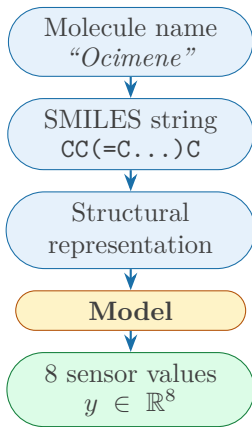
- ▶ Problem Statement
- ▶ The Electronic Nose
- ▶ Features Choice
- ▶ Results
- ▶ Graph Representation

The Problem

1 Problem Statement

Goal: given the identity of a molecule, predict how an electronic nose (e-nose) would respond to smelling it.

- Entirely a *representation* problem



Research Questions

1 Problem Statement

- How does the Aryballe e-nose work, and what does its sensing mechanism tell us about which molecular properties to use for prediction?
- Can we build a pipeline that automatically fetches SMILES from PubChem and computes a reproducible, filtered descriptors?
- Which descriptor family best captures the variation across these VOCs?
- How should descriptor selection be adapted when working with a very small molecule set, anticipating a larger dataset in future iterations?
- Does representing molecules as graphs preserve enough structural information to outperform handcrafted fingerprints on this regression task?

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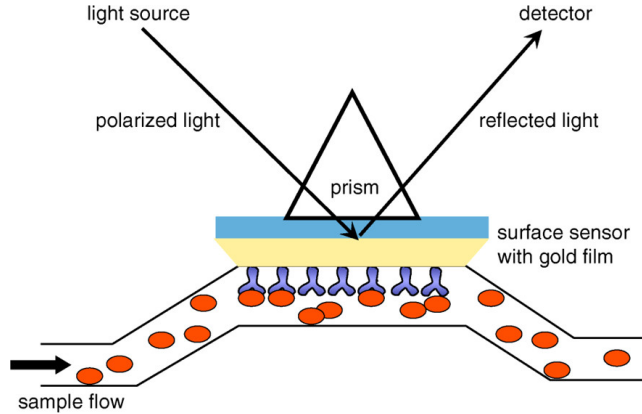
What Is an Electronic Nose?

2 The Electronic Nose

- Based on **Surface Plasmon Resonance imaging (SPRi)**
- **64 peptide sensors** on a gold film — each a short protein chain with a different 3-D shape
- A **laser** shines through a prism; a **camera** measures reflected light per sensor zone
- Molecules bind $\Rightarrow$ reflectivity **changes** $\Rightarrow$ signal recorded

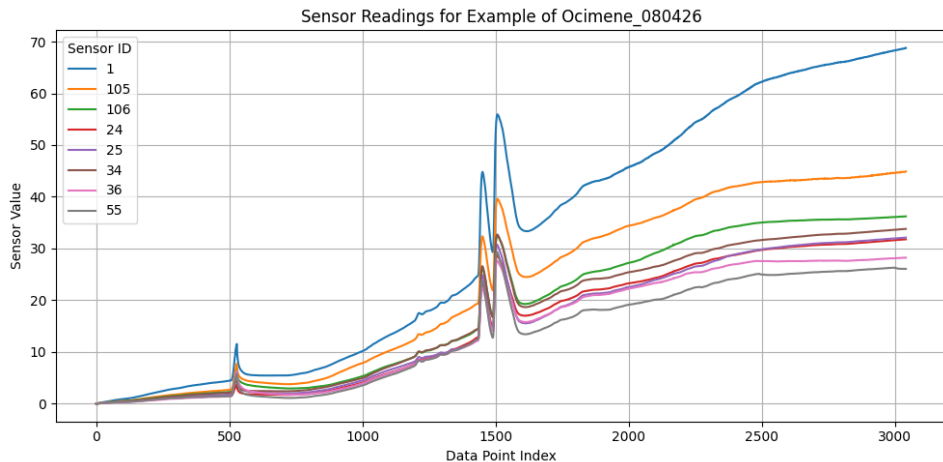
How a Measurement Is Acquired

2 The Electronic Nose



The Sensogram — Raw Sensor Output Over Time

2 The Electronic Nose



The Normalised Signatures — Dataset Overview

Mean normalised sensor values per molecule

Molecule	S1	S105	S106	S24	S25	S34	S36	S55
Ocimene	0.537	0.440	0.333	0.286	0.291	0.314	0.261	0.269
Δ^3 -Carene	0.418	0.452	0.330	0.327	0.325	0.363	0.274	0.302
Linalool	0.265	0.359	0.299	0.381	0.415	0.439	0.297	0.327
α -Pinene	0.170	0.236	0.249	0.562	0.380	0.430	0.308	0.331
(S)-Limonene	0.363	0.247	0.292	0.477	0.291	0.398	0.349	0.344
(R)-Limonene	0.400	0.250	0.308	0.259	0.291	0.432	0.440	0.383

The Smell Fingerprint — 8 Numbers per Molecule

2 The Electronic Nose

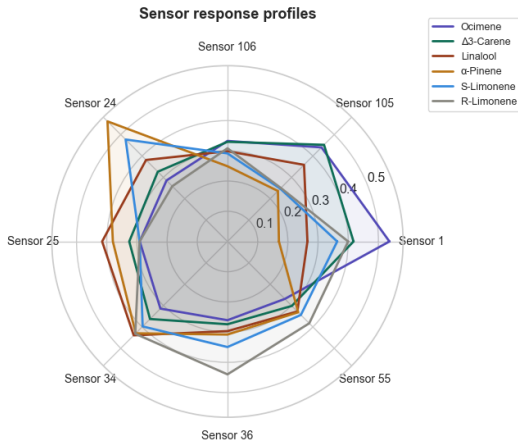


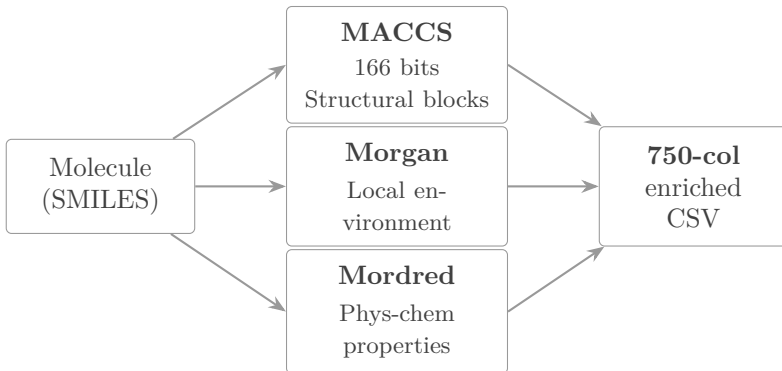
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3 Features Choice

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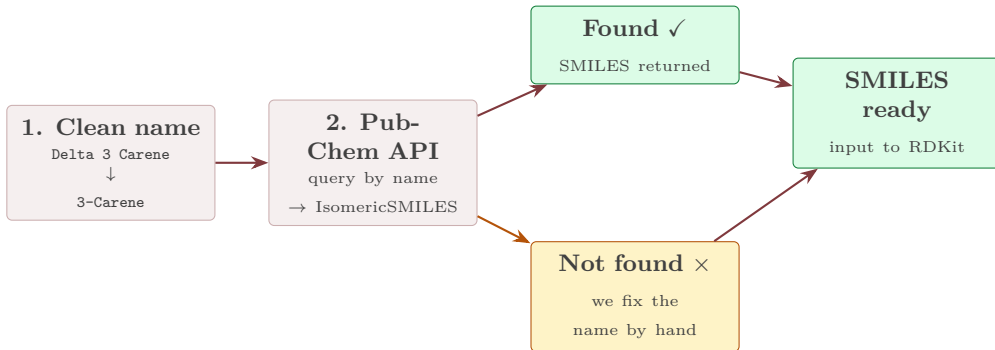
Feature Choice — 3 Complementary Families

3 Features Choice



From Molecule Name to SMILES

3 Features Choice



MACCS Keys

3 Features Choice

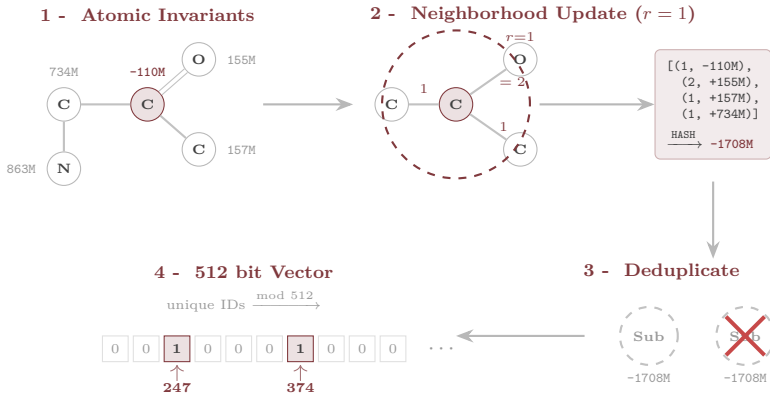
Has a ring? Has OH? Has N? Has
a double bond? ... **166 questions**



bit 141 active $\Rightarrow$ OH group present

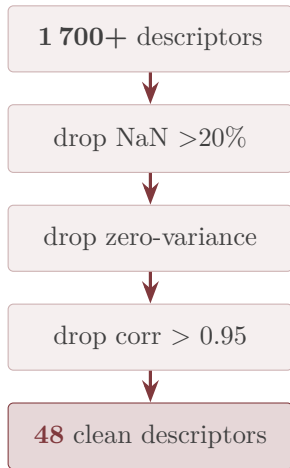
Morgan Fingerprints: ECFP Algorithm

3 Features Choice



Mordred Descriptors

3 Features Choice



What properties does it capture?

Volatility & logP

Air/Oil

MW · SLogP · VPCvt

Functional Groups



nAcid · nBase · nThiol

Steric / Size



VSA.EState · MV · Diameter

Binding Polarity

Donor Receptor



TPSA · nHBDon · nHBArc

Feature Extraction Pipeline

3 Features Choice



Mordred: 1700+ raw → NaN >20% → zero-variance → corr >0.95 → **48 clean descriptors**

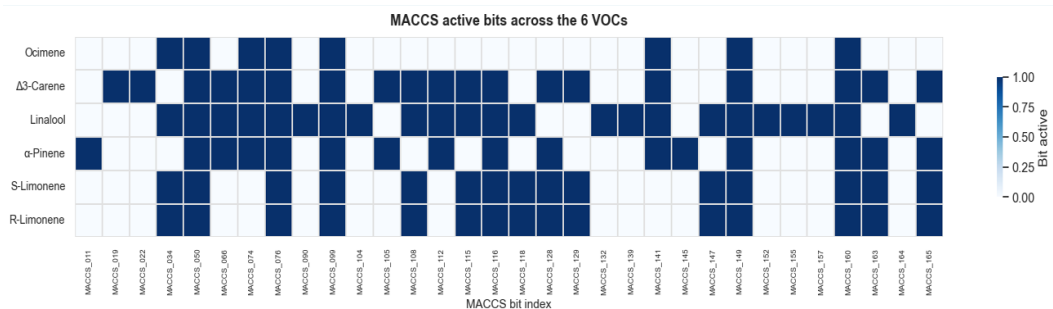
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4 Results

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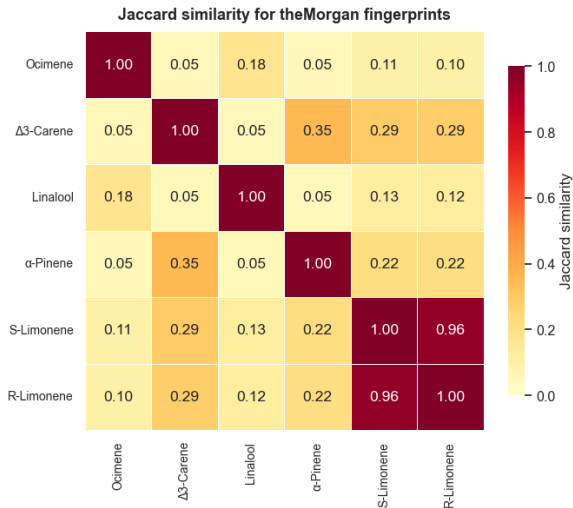
Result 1 : MACCS Active Bits

4 Results



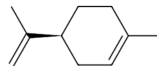
Result 2 : Jaccard Similarity for Morgan

4 Results

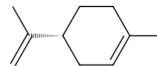


Jaccard similarity formula:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$



S-Limonene



R-Limonene

4 Results



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5 Graph Representation

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Why a Graph Representation?

5 Graph Representation

Extracted features

Morgan FP: 512 bits — fixed circular hash

MACCS keys: 167 bits — fixed vocabulary

Mordred: 48 scalars after filtering

Each method compresses structure with a **fixed, predetermined algorithm.**

Graph approach

Atoms as nodes — 9 features each

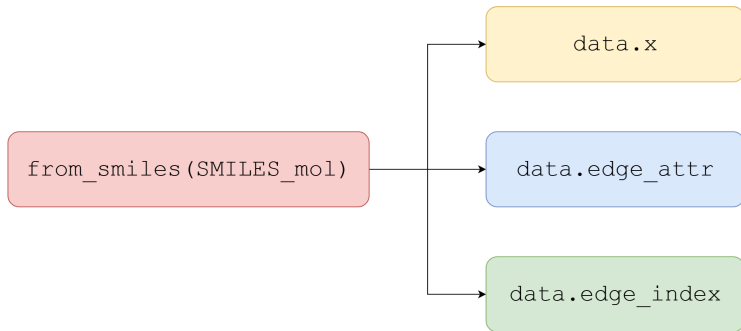
Bonds as edges — 5 features each

Full topology preserved — **no compression**

The model **learns** which structural patterns matter for this task.

Converting SMILES to a Graph

5 Graph Representation

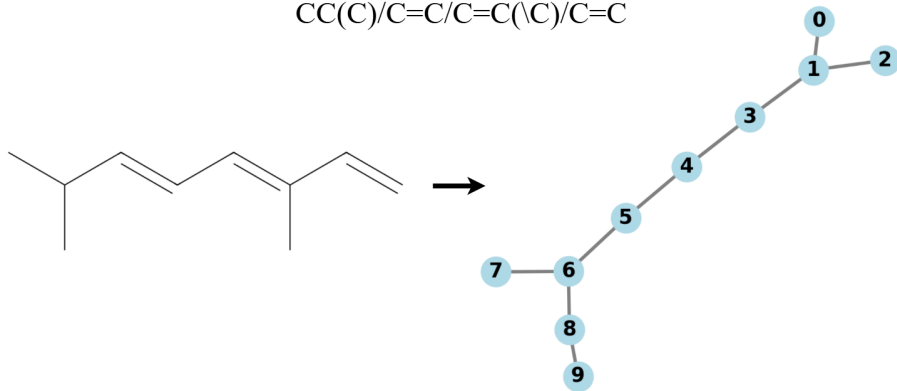


The Molecules as Graphs

5 Graph Representation

Ocimene

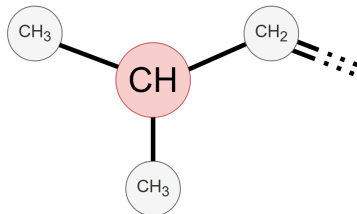
CC(C)/C=C/C=C\C/C=C



data.x — Node Feature Matrix

5 Graph Representation

Ocimene



[0] [1] [2] [3] [4] [5] [6] [7] [8]

6	0	3	0	1	0	2	0	0
---	---	---	---	---	---	---	---	---

#	Feature
---	---------

0	Atomic number (C=6, O=8)
---	--------------------------

1	Chirality ★
---	--------------------

2	Degree (bond count)
---	---------------------

3	Formal charge
---	---------------

4	Implicit H count
---	------------------

5	Radical electrons
---	-------------------

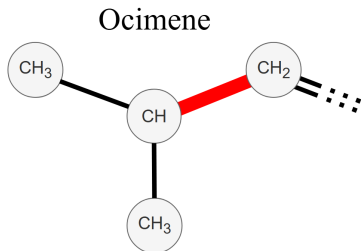
6	Hybridisation (SP/SP ² /SP ³)
---	--

7	Is aromatic? (always 0)
---	-------------------------

8	Is in ring? ★
---	----------------------

data.edge_attr + data.edge_index

5 Graph Representation



[0] [1] [2] [3] [4]

1	0	0	1	0
---	---	---	---	---

data.edge_attr — $[E \times 5]$

#	Feature
---	---------

0	Bond type (single/double/...)
---	-------------------------------

1	Stereo (Z/E, cis/trans)
---	-------------------------

2	Is aromatic?
---	--------------

3	Is conjugated?
---	----------------

4	Is in ring?
---	-------------

data.edge_index — $[2 \times E]$

The Complete Data Object

5 Graph Representation

Attribute	Shape	Content
<code>data.x</code>	$[atoms \times 9]$	Atom features
<code>data.edge_index</code>	$[2 \times E]$	Graph connectivity
<code>data.edge_attr</code>	$[E \times 5]$	Bond features
<code>data.y</code>	$[8]$	E-nose sensor values

Thank you for listening!
Your feedback will be highly appreciated!